COMPUTER ENGINEERING

Laboratorium
Multimedia dan Internet of Things
Departemen Teknik Komputer
*Institut Teknologi Sepuluh Nopember*

# Laporan Akhir
# Praktikum Jaringan Komputer

## Firewall & NAT

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# 1 Langkah-Langkah Percobaan

## Praktikum 1 Konfigurasi Cisco Switch

1. Masuk ke terminal Cisco Switch

   ```
   enable
   configure terminal
   ```

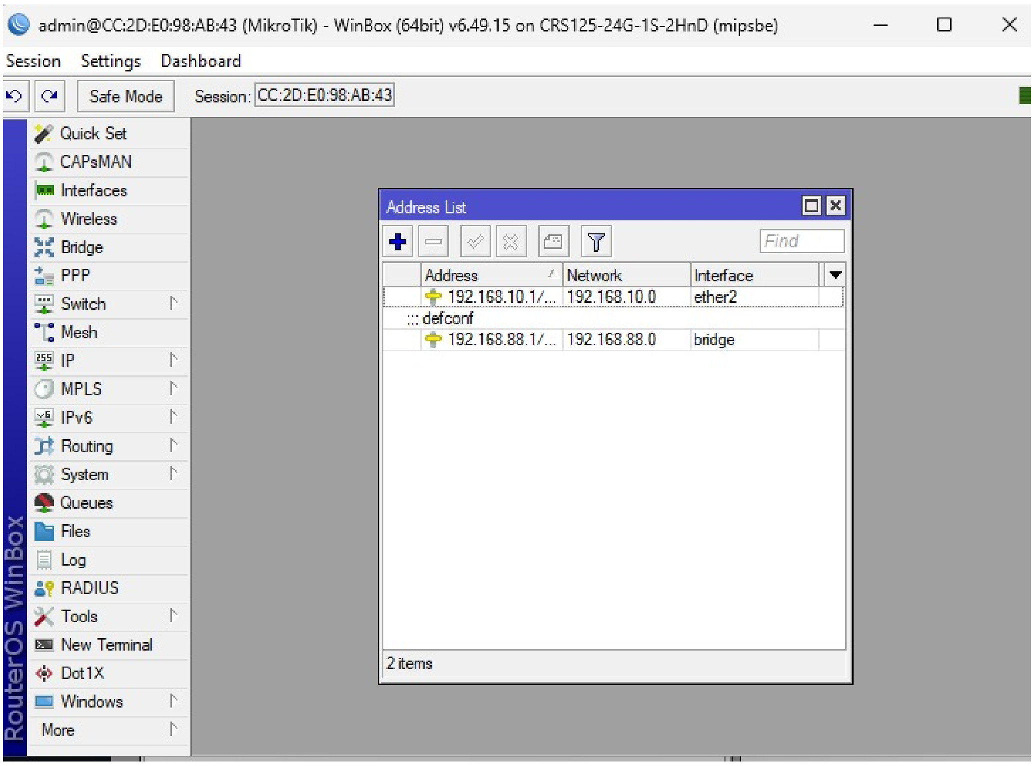
2. Membuat Vlan

   ```
   vlan 10
    name Finance
   exit

   vlan 20
    name HR
   exit

   vlan 30
    name IT
   exit

   vlan 40
    name Marketingg
   exit
   ```

   Verifikasi:

   ```
   show vlan brief
   ```

3. Pastikan vlan sudah dibuat dan statusnya aktif




**Gambar 1:** Hasil verifikasi pembuatan VLAN

4. Konfigurasi Access port untuk setiap vlan

```
conf t

interface gi0/2
 description PC-FINANCE-VLAN10
 switchport mode access
 switchport access vlan 10
 no shutdown
exit

interface gi0/1
 description PC-HR-VLAN20
 switchport mode access
 switchport access vlan 20
 no shutdown
exit

interface gi1/1
 description PC-IT-VLAN30
 switchport mode access
 switchport access vlan 30
 no shutdown
exit
```

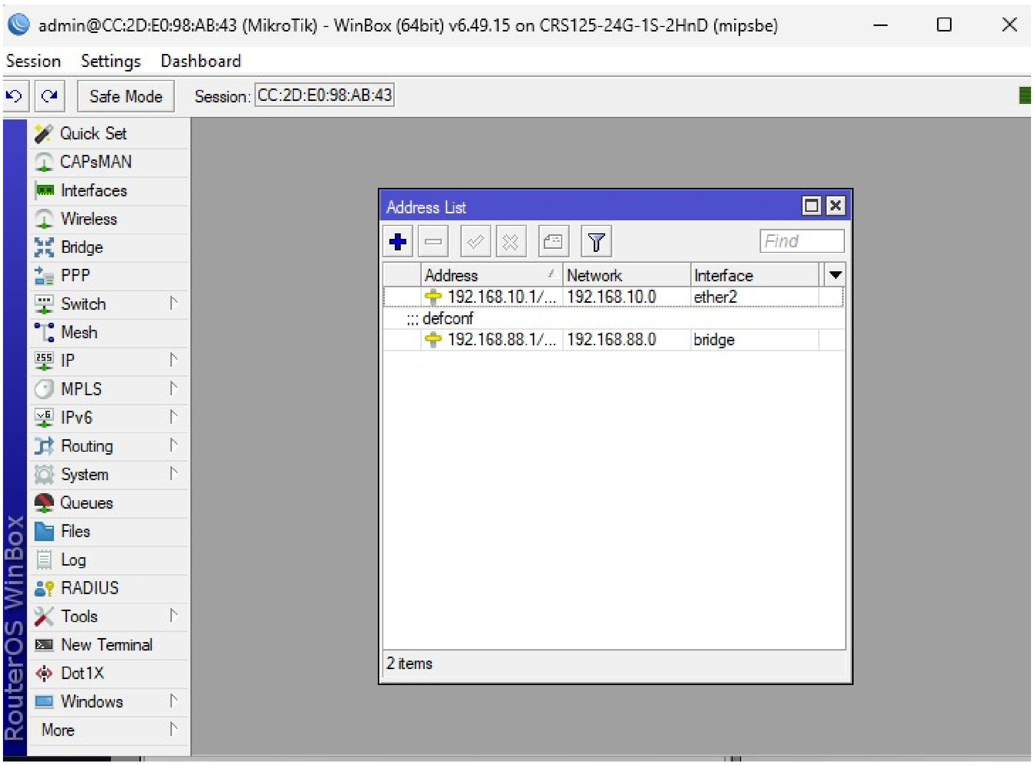
```
interface gi0/3
 description PC-MARKETING-VLAN40
 switchport mode access
 switchport access vlan 40
 no shutdown
exit
```

5. Konfigurasi Trunk ke Cisco Router

```
conf t

interface gi0/0
 switchport trunk encapsulation dot1q
 description TRUNK-TO-CISCO-ROUTER
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,40
 no shutdown
exit
```

Verifikasi:

```
show interfaces trunk
```




**Gambar 2:** Hasil verifikasi konfigurasi antarmuka trunk

Save semua konfigurasi dengan perintah:

```
write memory
```

6. Hasil yang diharapkan
   (a) Vlan 10, 20, 30, dan 40 sudah ada.
   (b) Trunk port sudah membawa vlan 10, 20, 30, dan 40.
   (c) Access port setiap vlan sudah sesuai.

## Praktikum 2 Konfigurasi Cisco Router

1. Masuk ke Cisco Router

```
enable
configure terminal
```

2. Konfigurasi Inteface Trunk ke Switch

```
interface gig0/2
 description TRUNK-TO-CISCO-SWITCH
 no ip address
 no shutdown
 exit
```

3. Konfigurasi Subinterface VLAN 10

```
interface gig0/2.10
 description GATEWAY-VLAN10-FINANCE
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
 no shutdown
exit
```

4. Konfigurasi Subinterface VLAN 20

```
interface gig0/2.20
 description GATEWAY-VLAN20-HR
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
 no shutdown
 exit
```

5. Konfigurasi Subinterface VLAN 30

```
interface gig0/2.30
 description GATEWAY-VLAN30-IT
 encapsulation dot1Q 30
 ip address 192.168.30.1 255.255.255.0
 no shutdown
 exit
```

6. Konfigurasi Subinterface VLAN 40

```
interface gig0/2.40
 description GATEWAY-VLAN40-MARKETING
 encapsulation dot1Q 40
 ip address 192.168.40.1 255.255.255.0
 no shutdown
 exit
```

7. Konfigurasi DHCP-Server untuk Vlan 10

```
ip dhcp excluded-address 192.168.10.1 192.168.10.10

ip dhcp pool VLAN10-FINANCE
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
 dns-server 8.8.8.8
exit
```

8. Konfigurasi DHCP-Server untuk Vlan 20

```
ip dhcp excluded-address 192.168.20.1 192.168.20.10

ip dhcp pool VLAN20-HR
 network 192.168.20.0 255.255.255.0
 default-router 192.168.20.1
 dns-server 8.8.8.8
exit
```

9. Konfigurasi IP Static pada VPCS untul vlan 30 dan 40 PC Vlan 30 :

```
ip 192.168.30.10/24 192.168.30.1
```

   PC Vlan 40 :

```
ip 192.168.40.10/24 192.168.40.1
```

   PC Vlan 10 dan 20 menggunakan dhcp:

```
ip dhcp
```

10. Verifikasi Cisco Router Cek interface:

```
show ip interface brief
```

Pastikan hasil adalah sebagai berikut:

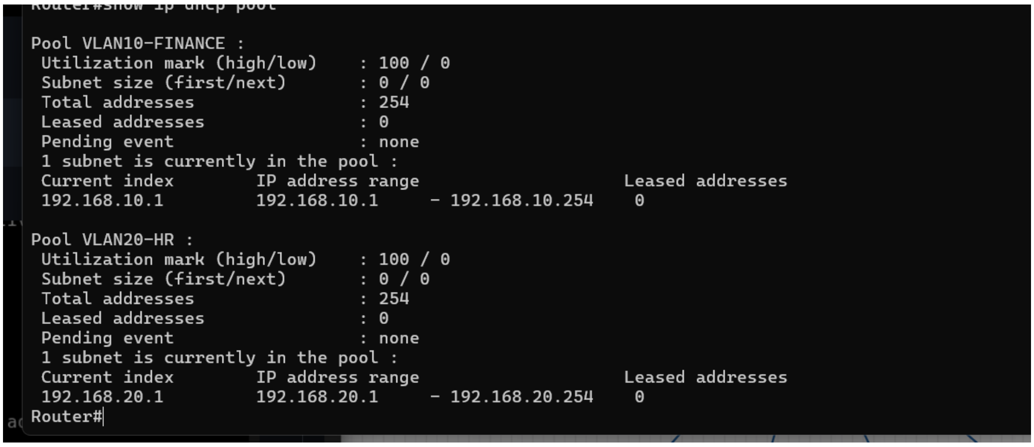
```
Router#show ip interface brief
Interface                  IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0         10.20.20.2      YES NVRAM  up                    up
GigabitEthernet0/1         192.168.10.1    YES NVRAM  up                    up
GigabitEthernet0/2         unassigned      YES NVRAM  administratively down down
GigabitEthernet0/3         unassigned      YES NVRAM  administratively down down
Router#
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 10.20.20.1 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 10.20.20.1
      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        10.20.20.0/30 is directly connected, GigabitEthernet0/0
L        10.20.20.2/32 is directly connected, GigabitEthernet0/0
      192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.10.0/24 is directly connected, GigabitEthernet0/1
L        192.168.10.1/32 is directly connected, GigabitEthernet0/1
Router#show running-config
Building configuration...

Current configuration : 2931 bytes
!
version 15.9
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname Router
!
boot-start-marker
boot-end-marker
!
!
!
no aaa new-model
!
!
!
mmi polling-interval 60
no mmi auto-configure
no mmi pvc
mmi snmp-timeout 180

Router#
```

**Gambar 3:** Hasil konfigurasi Cisco Router

Cek dhcp pool:

```
show ip dhcp pool
```

Pastikan hasil adalah sebagai berikut:




**Gambar 4:** Hasil verifikasi DHCP pool pada Cisco Router

Cek client DHCP yang mendapatkan IP:

```
show ip dhcp binding
```

11. Pengujian dari VPCS Masuk ke pc vlan 10 untuk request ip secara dhcp:

```
ip dhcp
```

Pastikan pc vlan 10 sudah dapat ip secara dhcp dengan command:

```
show ip
```

Dari pc vlan 10 ping gateway 192.168.10.1:

```
ping 192.168.10.1
```

Masuk ke pc vlan 20 untuk request ip secara dhcp:

```
ip dhcp
```

Pastikan pc vlan 20 sudah dapat ip secara dhcp dengan command:

```
show ip
```

Dari pc vlan 20 ping gateway 192.168.20.1:

```
ping 192.168.20.1
```

Dari pc vlan 30 ping gateway 192.168.30.1:

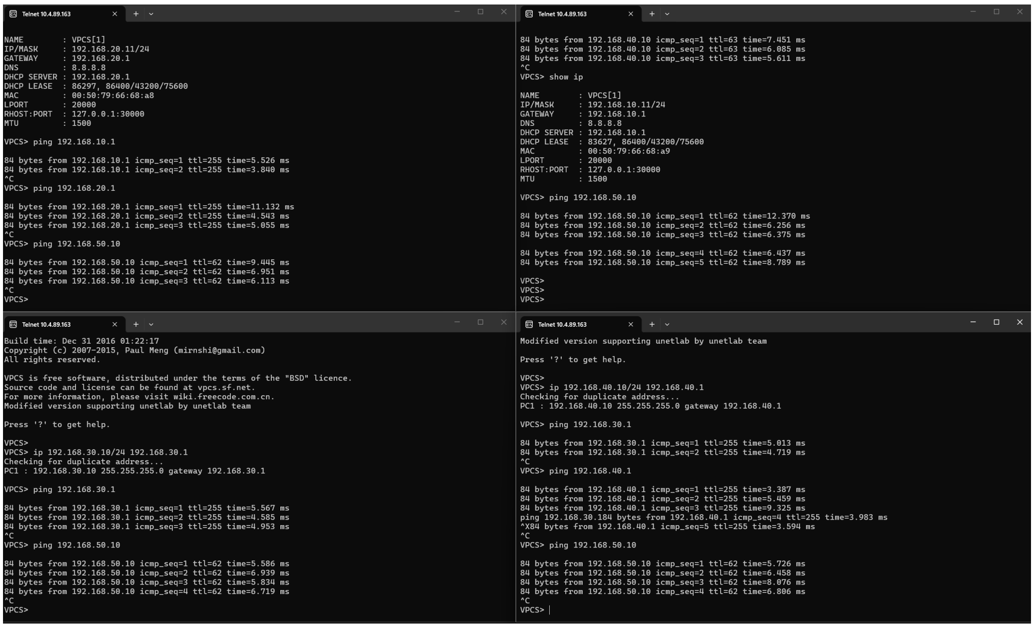
```
ping 192.168.30.1
```

Dari pc vlan 40 ping gateway 192.168.40.1:

```
ping 192.168.40.1
```

Uji ping antar-Vlan dari vlan 10:

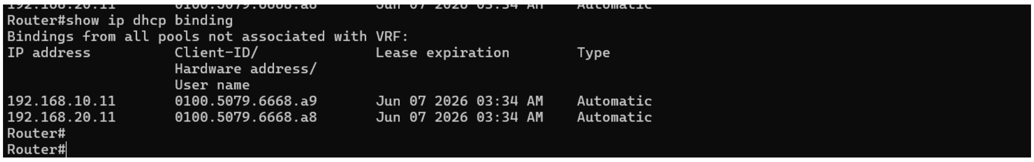
```
ping 192.168.20.11
ping 192.168.30.10
ping 192.168.40.10
```



**Gambar 5:** Hasil show ip dan ping antar-VLAN

Masuk ke cisco router lagi untuk melihat ip dhcp binding:

```
show ip dhcp binding
```




**Gambar 6:** Hasil verifikasi IP DHCP binding pada Cisco Router

Amati IP yang diberikan oleh router ke client sesuai dengan ip di pc vlan 10 dan 20.

## Praktikum 3 Konfigurasi Huawei Router dan IP Address Antarrouter

1. Konfigurasi Lan Huawei

```
system-view

interface Ethernet1/0/2
 description LAN-HUAWEI
 ip address 192.168.50.1 255.255.255.0
 undo shutdown
```

```
 quit

commit
```

2. Konfigurasi IP VPC Lan Huawei

   ```
   ip 192.168.50.10/24 192.168.50.1
   ```

   Verifikasi:

   ```
   show ip
   ```

   Testing ping gateway huawei:

   ```
   ping 192.168.50.1
   ```

   Pastikan hasilnya reply:

3. Konfigurasi IP link cisco to Huawei link 1 Cisco Router:

   ```
   enable
   configure terminal

   interface gi0/3
    description LINK1-TO-HUAWEI
    ip address 10.10.10.1 255.255.255.252
    no shutdown
   exit
   ```

   Huawei Router:

   ```
   system-view

   interface Ethernet1/0/0
    description LINK1-TO-CISCO
    ip address 10.10.10.2 255.255.255.252
    undo shutdown
    quit

   commit
   ```

4. Konfigurasi IP Link Cisco to Huawei Link 2 Cisco Router:

   ```
   interface gi0/0
    description LINK2-TO-HUAWEI
    ip address 10.10.11.1 255.255.255.252
    no shutdown
   exit
   ```

Huawei Router:

```
system-view

interface Ethernet1/0/4
 description LINK2-TO-CISCO
 ip address 10.10.11.2 255.255.255.252
 undo shutdown
 quit

commit
```

5. Konfigurasi IP link Huawei to Mikrotik Huawei Router:

```
system-view

interface Ethernet1/0/3
 description LINK-TO-MIKROTIK
 ip address 10.10.20.1 255.255.255.252
 undo shutdown
 quit

commit
```

Mikrotik:

```
/ip address add address=10.10.20.2/30 interface=ether1 comment=TO-HUAWEI
```

6. Konfigurasi IP link Cisco to Mikrotik Cisco Router:

```
interface gi0/1
 description LINK-TO-MIKROTIK
 ip address 10.10.30.1 255.255.255.252
 no shutdown
exit
```

Mikrotik:

```
/ip address add address=10.10.30.2/30 interface=ether2 comment=TO-CISCO
```

7. Verifikasi Interface Cisco Router:

```
show ip interface brief
```

```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.10.11.1      YES manual up                    up
GigabitEthernet0/1     10.10.30.1      YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  up                    up
GigabitEthernet0/2.10  192.168.10.1    YES manual up                    up
GigabitEthernet0/2.20  192.168.20.1    YES manual up                    up
GigabitEthernet0/2.30  192.168.30.1    YES manual up                    up
GigabitEthernet0/2.40  192.168.40.1    YES manual up                    up
GigabitEthernet0/3     10.10.10.1      YES manual up                    up
Router#
```

**Gambar 7:** Hasil verifikasi IP antarmuka pada Cisco Router (Praktikum 3)

Huawei Router:

```
display ip interface brief
```

```
Interface                         IP Address/Mask      Physical   Protocol  VPN
Ethernet1/0/0                     10.10.10.2/30        up         up        --
Ethernet1/0/0.4094                128.1.138.137/16     up         up        l3vpn
Ethernet1/0/1                     unassigned           up         down      --
Ethernet1/0/1.4094                128.1.138.137/16     up         up        l3vpn
Ethernet1/0/2                     192.168.50.1/24      up         up        --
Ethernet1/0/2.4094                128.1.138.137/16     up         up        l3vpn
Ethernet1/0/3                     10.10.20.1/30        up         up        --
Ethernet1/0/3.4094                128.1.138.137/16     up         up        l3vpn
Ethernet1/0/4                     10.10.11.2/30        up         up        --
Ethernet1/0/4.4094                128.1.138.137/16     up         up        l3vpn
Ethernet1/0/5                     unassigned           up         down      --
Ethernet1/0/5.4094                128.1.138.137/16     up         up        l3vpn
Ethernet1/0/6                     unassigned           up         down      --
Ethernet1/0/6.4094                128.1.138.137/16     up         up        l3vpn
Ethernet1/0/7                     unassigned           up         down      --
Ethernet1/0/7.4094                128.1.138.137/16     up         up        l3vpn
Ethernet1/0/8                     unassigned           up         down      --
Ethernet1/0/8.4094                128.1.138.137/16     up         up        l3vpn
Ethernet1/0/9                     unassigned           up         down      --
Ethernet1/0/9.4094                128.1.138.137/16     up         up        l3vpn
GigabitEthernet0/0/0              unassigned           up         down      --
LoopBack2147483647                128.1.138.137/16     up         up(s)     l3vpn
NULL0                             unassigned           up         up(s)     --

[~HUAWEI]
```

**Gambar 8:** Hasil verifikasi IP antarmuka pada Huawei Router

Mikrotik:

```
/ip address print
```

```
[admin@MikroTik] > /ip address print
Flags: X - disabled, I - invalid, D - dynamic
 #   ADDRESS            NETWORK         INTERFACE
 0   ;;; TO-HUAWEI
     10.10.20.2/30      10.10.20.0      ether1
 1   ;;; TO-CISCO
     10.10.30.2/30      10.10.30.0      ether2
```

**Gambar 9:** Hasil verifikasi IP address pada MikroTik

8. Verifikasi ping Antar Router Dari Cisco Router:

```
ping 10.10.10.2
ping 10.10.11.2
ping 10.10.30.2
```

```
*Jun  6 04:01:15.390: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to upping 10.10.
10.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/3/5 ms
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/3/5 ms
Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/5 ms
Router#
```

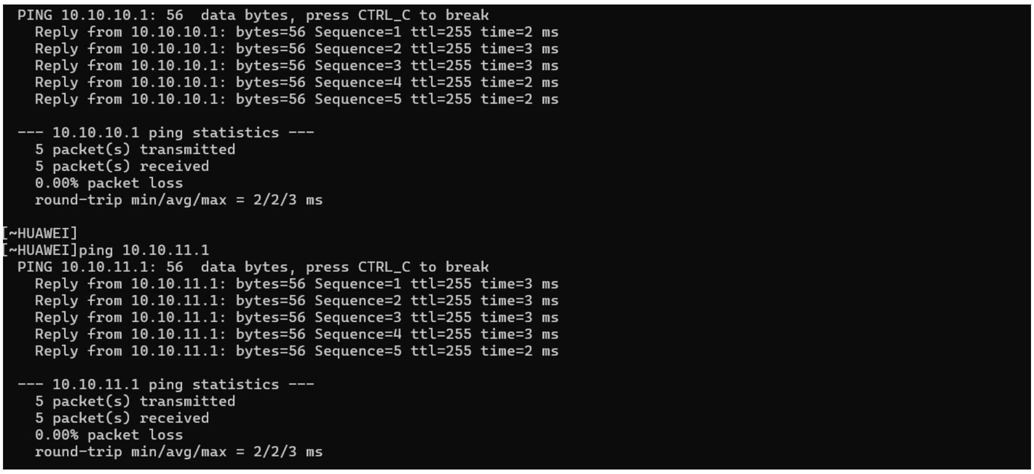
**Gambar 10:** Hasil ping dari Cisco Router ke router tetangga

Dari Router Huawei:

```
ping 10.10.10.1
ping 10.10.11.1
ping 10.10.20.2
```

```
  Reply from 10.10.10.1: bytes=56 Sequence=4 ttl=255 time=2 ms
  Reply from 10.10.10.1: bytes=56 Sequence=5 ttl=255 time=2 ms

 --- 10.10.10.1 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 2/2/3 ms

~HUAWEI]
~HUAWEI]ping 10.10.11.1
 PING 10.10.11.1: 56  data bytes, press CTRL_C to break
  Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=3 ms
  Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=3 ms
  Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=3 ms
  Reply from 10.10.11.1: bytes=56 Sequence=4 ttl=255 time=3 ms
  Reply from 10.10.11.1: bytes=56 Sequence=5 ttl=255 time=2 ms

 --- 10.10.11.1 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 2/2/3 ms

~HUAWEI]
~HUAWEI]ping 10.10.20.2
 PING 10.10.20.2: 56  data bytes, press CTRL_C to break
  Reply from 10.10.20.2: bytes=56 Sequence=1 ttl=64 time=1 ms
  Reply from 10.10.20.2: bytes=56 Sequence=2 ttl=64 time=1 ms
  Reply from 10.10.20.2: bytes=56 Sequence=3 ttl=64 time=1 ms
  Reply from 10.10.20.2: bytes=56 Sequence=4 ttl=64 time=2 ms
  Reply from 10.10.20.2: bytes=56 Sequence=5 ttl=64 time=2 ms

 --- 10.10.20.2 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 1/1/2 ms
```

**Gambar 11:** Hasil ping dari Huawei Router ke router tetangga (1)

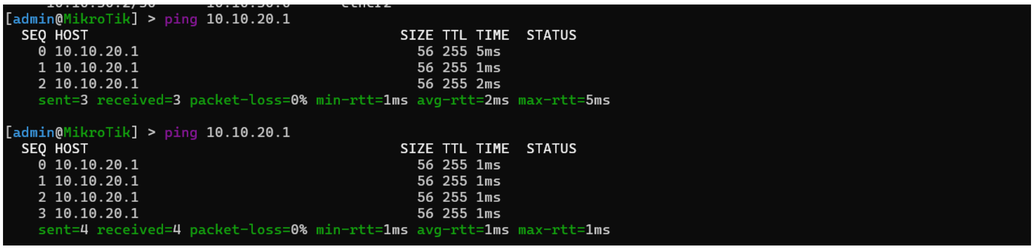


```
 PING 10.10.10.1: 56  data bytes, press CTRL_C to break
   Reply from 10.10.10.1: bytes=56 Sequence=1 ttl=255 time=2 ms
   Reply from 10.10.10.1: bytes=56 Sequence=2 ttl=255 time=3 ms
   Reply from 10.10.10.1: bytes=56 Sequence=3 ttl=255 time=3 ms
   Reply from 10.10.10.1: bytes=56 Sequence=4 ttl=255 time=2 ms
   Reply from 10.10.10.1: bytes=56 Sequence=5 ttl=255 time=2 ms

 --- 10.10.10.1 ping statistics ---
   5 packet(s) transmitted
   5 packet(s) received
   0.00% packet loss
   round-trip min/avg/max = 2/2/3 ms

[~HUAWEI]
[~HUAWEI]ping 10.10.11.1
 PING 10.10.11.1: 56  data bytes, press CTRL_C to break
   Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=3 ms
   Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=3 ms
   Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=3 ms
   Reply from 10.10.11.1: bytes=56 Sequence=4 ttl=255 time=3 ms
   Reply from 10.10.11.1: bytes=56 Sequence=5 ttl=255 time=2 ms

 --- 10.10.11.1 ping statistics ---
   5 packet(s) transmitted
   5 packet(s) received
   0.00% packet loss
   round-trip min/avg/max = 2/2/3 ms
```

**Gambar 12:** Hasil ping dari Huawei Router ke router tetangga (2)

Dari Router Mikrotik:

```
ping 10.10.20.1
ping 10.10.30.1
```



```
[admin@MikroTik] > ping 10.10.20.1
  SEQ HOST                                     SIZE TTL TIME  STATUS
    0 10.10.20.1                                 56 255 5ms
    1 10.10.20.1                                 56 255 1ms
    2 10.10.20.1                                 56 255 2ms
    sent=3 received=3 packet-loss=0% min-rtt=1ms avg-rtt=2ms max-rtt=5ms

[admin@MikroTik] > ping 10.10.20.1
  SEQ HOST                                     SIZE TTL TIME  STATUS
    0 10.10.20.1                                 56 255 1ms
    1 10.10.20.1                                 56 255 1ms
    2 10.10.20.1                                 56 255 1ms
    3 10.10.20.1                                 56 255 1ms
    sent=4 received=4 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=1ms
```

**Gambar 13:** Hasil ping dari MikroTik ke router tetangga

## Praktikum 4 Konfigurasi OSPF Multivendor dan OSPF Cost

1. Konsep routing ospf pada topologi Pada topologi ini, cisco router dan huawei memiliki 2 jalur utama. Lalu, mikrotik digunakan sebagai jalur cadangan jika 2 link utama tidak berfungsi.
2. Network yang diiklankan ke ospf Perhatikan tabel routing yang sesuai untuk masing-masing router.
3. Konfigurasi ospf pada Cisco Router Masuk ke Cisco Router:

   ```
   enable
   configure terminal
   ```

   Konfigurasi OSPF:

   ```
   router ospf 1
    router-id 1.1.1.1
    network 10.10.10.0 0.0.0.3 area 0
    network 10.10.11.0 0.0.0.3 area 0
   ```

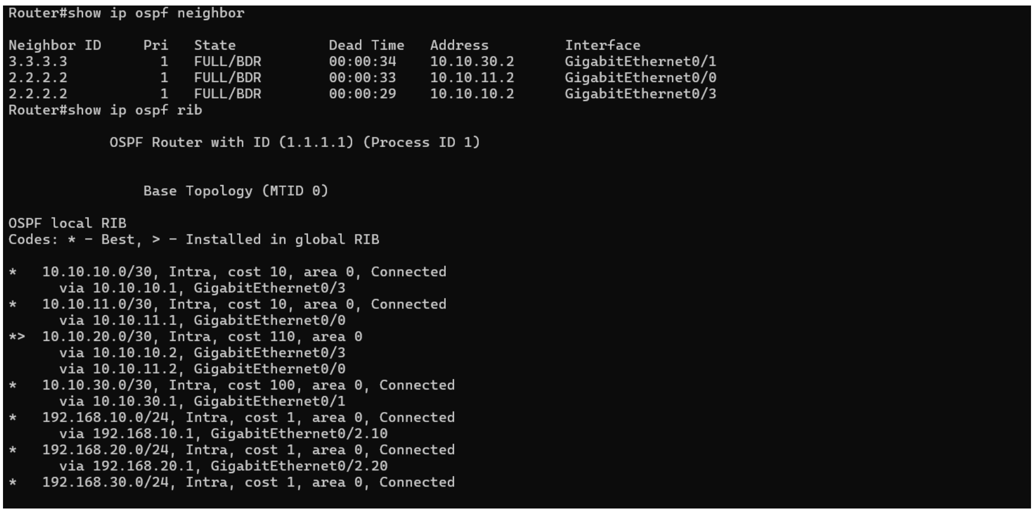
```
 network 10.10.30.0 0.0.0.3 area 0
 network 192.168.10.0 0.0.0.255 area 0
 network 192.168.20.0 0.0.0.255 area 0
 network 192.168.30.0 0.0.0.255 area 0
 network 192.168.40.0 0.0.0.255 area 0
exit
```

Atur OSPF cost pada interface Cisco:

```
interface gi0/3
 description LINK1-TO-HUAWEI
 ip ospf cost 10
exit

interface gi0/0
 description LINK2-TO-HUAWEI
 ip ospf cost 10
exit

interface gi0/1
 description LINK-TO-MIKROTIK
 ip ospf cost 100
exit
```

Verifikasi OSPF Cisco:

```
show ip ospf neighbor
show ip route ospf
show ip protocols
```



```
Router#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
3.3.3.3           1   FULL/BDR        00:00:34    10.10.30.2      GigabitEthernet0/1
2.2.2.2           1   FULL/BDR        00:00:33    10.10.11.2      GigabitEthernet0/0
2.2.2.2           1   FULL/BDR        00:00:29    10.10.10.2      GigabitEthernet0/3
Router#show ip ospf rib

            OSPF Router with ID (1.1.1.1) (Process ID 1)


                Base Topology (MTID 0)

OSPF local RIB
Codes: * - Best, > - Installed in global RIB

*   10.10.10.0/30, Intra, cost 10, area 0, Connected
      via 10.10.10.1, GigabitEthernet0/3
*   10.10.11.0/30, Intra, cost 10, area 0, Connected
      via 10.10.11.1, GigabitEthernet0/0
*>  10.10.20.0/30, Intra, cost 110, area 0
      via 10.10.10.2, GigabitEthernet0/3
      via 10.10.11.2, GigabitEthernet0/0
*   10.10.30.0/30, Intra, cost 100, area 0, Connected
      via 10.10.30.1, GigabitEthernet0/1
*   192.168.10.0/24, Intra, cost 1, area 0, Connected
      via 192.168.10.1, GigabitEthernet0/2.10
*   192.168.20.0/24, Intra, cost 1, area 0, Connected
      via 192.168.20.1, GigabitEthernet0/2.20
*   192.168.30.0/24, Intra, cost 1, area 0, Connected
```

**Gambar 14:** Verifikasi OSPF protocol dan network pada Cisco Router

4. Konfigurasi OSPF pada Huawei Router Masuk ke huawei router:

```
system-view
```

Konfigurasi OSPF:

```
ospf 1 router-id 2.2.2.2
 area 0.0.0.0
  network 10.10.10.0 0.0.0.3
  network 10.10.11.0 0.0.0.3
  network 10.10.20.0 0.0.0.3
  network 192.168.50.0 0.0.0.255
 quit
quit
```

Atur OSPF cost pada interface huawei:

```
interface Ethernet1/0/0
 description LINK1-TO-CISCO
 ospf cost 10
 quit

interface Ethernet1/0/4
 description LINK2-TO-CISCO
 ospf cost 10
 quit

interface Ethernet1/0/3
 description LINK-TO-MIKROTIK
 ospf cost 100
 quit

commit
```

Verifikasi OSPF Huawei:

```
display ospf peer
```

```
[~HUAWEI]display ospf peer

(M) Indicates MADJ neighbor


          OSPF Process 1 with Router ID 2.2.2.2
                Neighbors

 Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
 Router ID: 1.1.1.1              Address: 10.10.10.1
   State: Full           Mode:Nbr is Slave      Priority: 1
   DR: 10.10.10.1        BDR: 10.10.10.2        MTU: 1500
   Dead timer due in  37  sec
   Retrans timer interval: 5
   Neighbor is up for 00h00m09s
   Neighbor Up Time : 2026-06-06 04:07:43
   Authentication Sequence: [ 0 ]

 Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
 Router ID: 1.1.1.1              Address: 10.10.11.1
   State: Full           Mode:Nbr is Slave      Priority: 1
   DR: 10.10.11.1        BDR: 10.10.11.2        MTU: 1500
   Dead timer due in  37  sec
   Retrans timer interval: 5
   Neighbor is up for 00h00m09s
   Neighbor Up Time : 2026-06-06 04:07:43
   Authentication Sequence: [ 0 ]
```

**Gambar 15:** Verifikasi OSPF peer pada Huawei Router

```
display ip routing-table protocol ospf
```

```
[~HUAWEI]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
------------------------------------------------------------------------------
_public_ Routing Table : OSPF
         Destinations : 9        Routes : 14

OSPF routing table status : <Active>
         Destinations : 5        Routes : 10

Destination/Mask    Proto   Pre  Cost        Flags NextHop         Interface

     10.10.30.0/30  OSPF    10   110           D   10.10.11.1      Ethernet1/0/4
                    OSPF    10   110           D   10.10.10.1      Ethernet1/0/0
   192.168.10.0/24  OSPF    10   11            D   10.10.11.1      Ethernet1/0/4
                    OSPF    10   11            D   10.10.10.1      Ethernet1/0/0
   192.168.20.0/24  OSPF    10   11            D   10.10.11.1      Ethernet1/0/4
                    OSPF    10   11            D   10.10.10.1      Ethernet1/0/0
   192.168.30.0/24  OSPF    10   11            D   10.10.11.1      Ethernet1/0/4
                    OSPF    10   11            D   10.10.10.1      Ethernet1/0/0
   192.168.40.0/24  OSPF    10   11            D   10.10.11.1      Ethernet1/0/4
                    OSPF    10   11            D   10.10.10.1      Ethernet1/0/0

OSPF routing table status : <Inactive>
         Destinations : 4        Routes : 4

Destination/Mask    Proto   Pre  Cost        Flags NextHop         Interface

     10.10.10.0/30  OSPF    10   10                10.10.10.2      Ethernet1/0/0
     10.10.11.0/30  OSPF    10   10                10.10.11.2      Ethernet1/0/4
     10.10.20.0/30  OSPF    10   100               10.10.20.1      Ethernet1/0/3
   192.168.50.0/24  OSPF    10   1                 192.168.50.1    Ethernet1/0/2
[~HUAWEI]
```

**Gambar 16:** Tabel routing OSPF pada Huawei Router

```
display ospf routing
```

```
[~HUAWEI]display ospf routing

          OSPF Process 1 with Router ID 2.2.2.2
                   Routing Tables

 Routing for Network
 Destination        Cost     Type       NextHop         AdvRouter       Area
 10.10.10.0/30      10       Direct     10.10.10.2      2.2.2.2         0.0.0.0
 10.10.11.0/30      10       Direct     10.10.11.2      2.2.2.2         0.0.0.0
 10.10.20.0/30      100      Direct     10.10.20.1      2.2.2.2         0.0.0.0
 10.10.30.0/30      110      Stub       10.10.11.1      1.1.1.1         0.0.0.0
 10.10.30.0/30      110      Stub       10.10.10.1      1.1.1.1         0.0.0.0
 192.168.10.0/24    11       Stub       10.10.11.1      1.1.1.1         0.0.0.0
 192.168.10.0/24    11       Stub       10.10.10.1      1.1.1.1         0.0.0.0
 192.168.20.0/24    11       Stub       10.10.11.1      1.1.1.1         0.0.0.0
 192.168.20.0/24    11       Stub       10.10.10.1      1.1.1.1         0.0.0.0
 192.168.30.0/24    11       Stub       10.10.11.1      1.1.1.1         0.0.0.0
 192.168.30.0/24    11       Stub       10.10.10.1      1.1.1.1         0.0.0.0
 192.168.40.0/24    11       Stub       10.10.11.1      1.1.1.1         0.0.0.0
 192.168.40.0/24    11       Stub       10.10.10.1      1.1.1.1         0.0.0.0
 192.168.50.0/24    1        Direct     192.168.50.1    2.2.2.2         0.0.0.0

 Total Nets: 9
 Intra Area: 9  Inter Area: 0  ASE: 0  NSSA: 0
  ---- More ----
```

**Gambar 17:** Detail OSPF routing pada Huawei Router

5. Konfigurasi OSPF pada MikroTik Masuk ke terminal Mikrotik:

   ```
   /routing ospf instance set default router-id=3.3.3.3
   ```

   Tambahkan network OSPF:

   ```
   /routing ospf network add network=10.10.20.0/30 area=backbone
   /routing ospf network add network=10.10.30.0/30 area=backbone
   ```

   Atur cost interface mikrotik menjadi jalur bakcup:

   ```
   /routing ospf interface add interface=ether1 cost=100
   /routing ospf interface add interface=ether2 cost=100
   ```

   Verifikasi Mikrotik:

   ```
   /routing ospf neighbor print
   ```

```
[admin@MikroTik] > /routing ospf neighbor print
 0 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1 dr-address=10.10.20.1
   backup-dr-address=10.10.20.2 state="Full" state-changes=5 ls-retransmits=0 ls-requests=0 db-summaries=0
   adjacency=3m5s

 1 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1 dr-address=10.10.30.1
   backup-dr-address=10.10.30.2 state="Full" state-changes=5 ls-retransmits=0 ls-requests=0 db-summaries=0
   adjacency=3m12s
[admin@MikroTik] >
```

**Gambar 18:** Verifikasi OSPF neighbor pada MikroTik

   Cek routing table OSPF:

   ```
   /ip route print where ospf
   ```

```
[admin@MikroTik] > /routing ospf neighbor print
 0 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1 dr-address=10.10.20.1
   backup-dr-address=0.0.0.0 state="Full" state-changes=5 ls-retransmits=1 ls-requests=0 db-summaries=0 adjacency=4s

 1 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1 dr-address=10.10.30.1
   backup-dr-address=10.10.30.2 state="Full" state-changes=5 ls-retransmits=0 ls-requests=0 db-summaries=0
   adjacency=11s
[admin@MikroTik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic, C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
 #      DST-ADDRESS        PREF-SRC        GATEWAY            DISTANCE
 0 ADo  10.10.10.0/30                      10.10.30.1              110
                                           10.10.20.1
 1 ADo  10.10.11.0/30                      10.10.30.1              110
                                           10.10.20.1
 2 ADo  192.168.10.0/24                    10.10.30.1              110
 3 ADo  192.168.20.0/24                    10.10.30.1              110
 4 ADo  192.168.30.0/24                    10.10.30.1              110
 5 ADo  192.168.40.0/24                    10.10.30.1              110
 6 ADo  192.168.50.0/24                    10.10.20.1              110
[admin@MikroTik] > |
```

**Gambar 19:** Tabel routing OSPF pada MikroTik

6. Verifikasi OSPF Neighbor Cisco Router:

```
show ip ospf neighbor
```

```
Router#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
3.3.3.3           1   FULL/BDR        00:00:33    10.10.30.2      GigabitEthernet0/1
2.2.2.2           1   FULL/BDR        00:00:31    10.10.11.2      GigabitEthernet0/0
2.2.2.2           1   FULL/BDR        00:00:32    10.10.10.2      GigabitEthernet0/3
Router#
```

**Gambar 20:** Verifikasi OSPF neighbor pada Cisco Router

Huawei Router:

```
display ospf peer
```

```
[~HUAWEI]display ospf peer

(M) Indicates MADJ neighbor


          OSPF Process 1 with Router ID 2.2.2.2
                Neighbors

 Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
 Router ID: 1.1.1.1              Address: 10.10.10.1
   State: Full          Mode:Nbr is Slave      Priority: 1
   DR: 10.10.10.1       BDR: 10.10.10.2        MTU: 1500
   Dead timer due in  37  sec
   Retrans timer interval: 5
   Neighbor is up for 00h00m09s
   Neighbor Up Time : 2026-06-06 04:07:43
   Authentication Sequence: [ 0 ]

 Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
 Router ID: 1.1.1.1              Address: 10.10.11.1
   State: Full          Mode:Nbr is Slave      Priority: 1
   DR: 10.10.11.1       BDR: 10.10.11.2        MTU: 1500
   Dead timer due in  37  sec
   Retrans timer interval: 5
   Neighbor is up for 00h00m09s
   Neighbor Up Time : 2026-06-06 04:07:43
   Authentication Sequence: [ 0 ]
```

**Gambar 21:** Verifikasi OSPF peer pada Huawei Router (lanjutan)

Mikrotik:

```
/routing ospf neighbor print
```

```
[admin@MikroTik] > /routing ospf neighbor print
 0 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1 dr-address=10.10.20.1
   backup-dr-address=10.10.20.2 state="Full" state-changes=5 ls-retransmits=0 ls-requests=0 db-summaries=0
   adjacency=3m5s

 1 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1 dr-address=10.10.30.1
   backup-dr-address=10.10.30.2 state="Full" state-changes=5 ls-retransmits=0 ls-requests=0 db-summaries=0
   adjacency=3m12s
[admin@MikroTik] > |
```

**Gambar 22:** Verifikasi OSPF neighbor pada MikroTik (lanjutan)

7. Verifikasi Routing Table Cisco Router:

```
show ip route 192.168.50.0
```

```
show ip route ospf
```

Huawei Router:

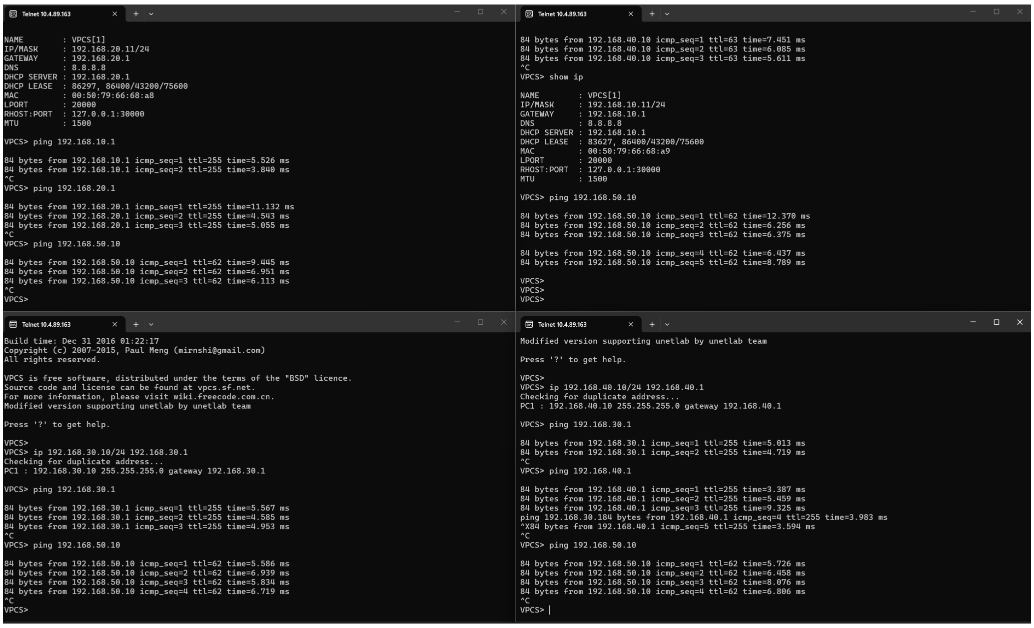
```
display ip routing-table protocol ospf
```

```
[~HUAWEI]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
------------------------------------------------------------------------------
_public_ Routing Table : OSPF
         Destinations : 9        Routes : 14

OSPF routing table status : <Active>
         Destinations : 5        Routes : 10

Destination/Mask    Proto   Pre  Cost        Flags NextHop         Interface

     10.10.30.0/30  OSPF    10   110           D   10.10.11.1      Ethernet1/0/4
                    OSPF    10   110           D   10.10.10.1      Ethernet1/0/0
   192.168.10.0/24  OSPF    10   11            D   10.10.11.1      Ethernet1/0/4
                    OSPF    10   11            D   10.10.10.1      Ethernet1/0/0
   192.168.20.0/24  OSPF    10   11            D   10.10.11.1      Ethernet1/0/4
                    OSPF    10   11            D   10.10.10.1      Ethernet1/0/0
   192.168.30.0/24  OSPF    10   11            D   10.10.11.1      Ethernet1/0/4
                    OSPF    10   11            D   10.10.10.1      Ethernet1/0/0
   192.168.40.0/24  OSPF    10   11            D   10.10.11.1      Ethernet1/0/4
                    OSPF    10   11            D   10.10.10.1      Ethernet1/0/0

OSPF routing table status : <Inactive>
         Destinations : 4        Routes : 4

Destination/Mask    Proto   Pre  Cost        Flags NextHop         Interface

     10.10.10.0/30  OSPF    10   10                10.10.10.2      Ethernet1/0/0
     10.10.11.0/30  OSPF    10   10                10.10.11.2      Ethernet1/0/4
     10.10.20.0/30  OSPF    10   100               10.10.20.1      Ethernet1/0/3
   192.168.50.0/24  OSPF    10   1                 192.168.50.1    Ethernet1/0/2
[~HUAWEI]|
```

**Gambar 23:** Keseluruhan tabel routing OSPF pada Huawei Router

Mikrotik:

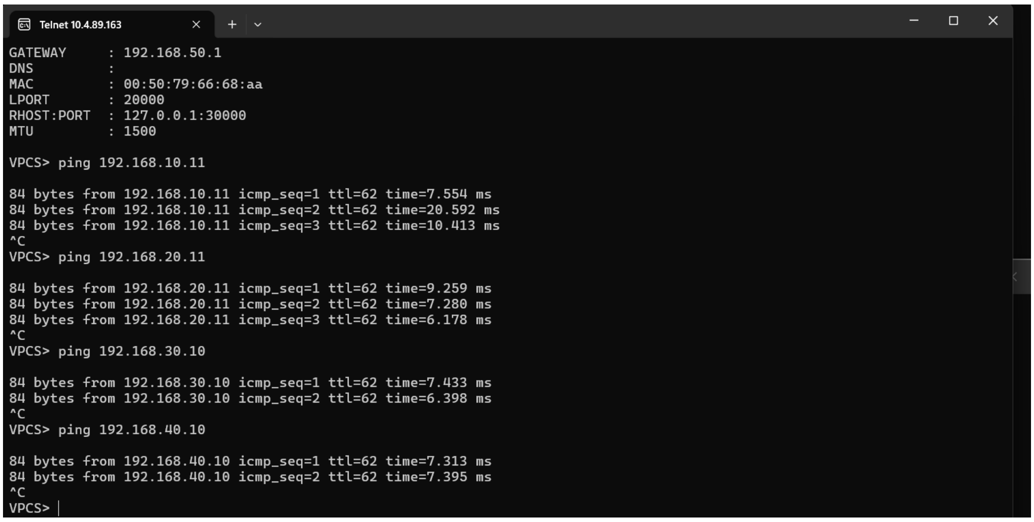
```
/ip route print where ospf
    \item Verifikasi Koneksi end-to-end
    Dari pc vlan 10, 20, 30, 40:
\begin{verbatim}
ping 192.168.50.10
```



**Gambar 24:** Hasil ping end-to-end dari PC VLAN ke LAN Huawei

Dari pc lan huawei ke ip vlan cisco:

```
ping 192.168.10.11
ping 192.168.20.11
ping 192.168.30.10
ping 192.168.40.10
```

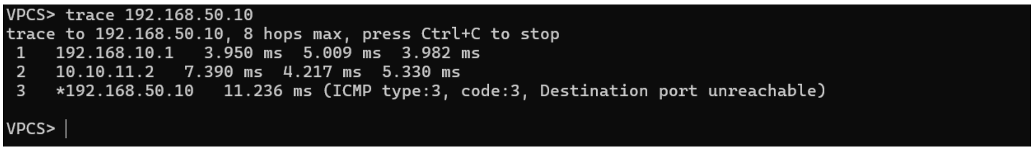



**Gambar 25:** Hasil ping end-to-end dari PC LAN Huawei ke seluruh VLAN

## Praktikum 5 Pengujian Failover OSPF

1. Kondisi normal: Dua link cisco-huawei aktif

```
show ip route 192.168.50.0
trace 192.168.50.10
```

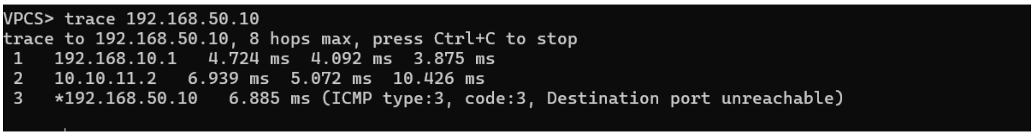


```
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1   192.168.10.1   3.950 ms  5.009 ms  3.982 ms
 2   10.10.11.2   7.390 ms  4.217 ms  5.330 ms
 3   *192.168.50.10   11.236 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS> |
```

**Gambar 26:** Hasil trace route pada kondisi normal (dua link utama aktif)

2. Kondisi 1 link cisco-huawei mati

```
configure terminal
interface gi0/3
 shutdown
exit

show ip route 192.168.50.0
trace 192.168.50.10
```

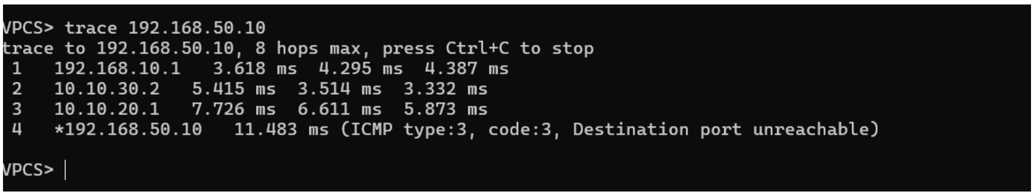


```
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1   192.168.10.1   4.724 ms  4.092 ms  3.875 ms
 2   10.10.11.2   6.939 ms  5.072 ms  10.426 ms
 3   *192.168.50.10   6.885 ms (ICMP type:3, code:3, Destination port unreachable)
```

**Gambar 27:** Hasil trace route saat satu link utama Cisco-Huawei mati

3. Kondisi dua link cisco-huawei mati

```
configure terminal
interface gi0/0
 shutdown
exit

show ip route 192.168.50.0
trace 192.168.50.10
```



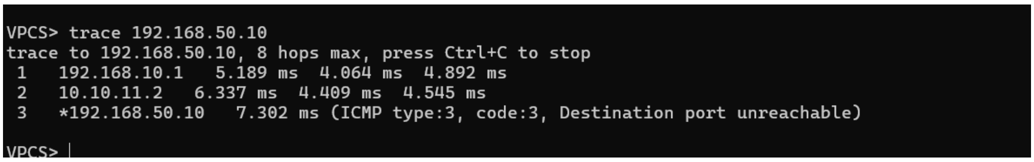
```
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1   192.168.10.1   3.618 ms  4.295 ms  4.387 ms
 2   10.10.30.2   5.415 ms  3.514 ms  3.332 ms
 3   10.10.20.1   7.726 ms  6.611 ms  5.873 ms
 4   *192.168.50.10   11.483 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS> |
```

**Gambar 28:** Hasil trace route saat dua link utama mati (berpindah ke jalur backup MikroTik)

4. Mengaktifkan kembali dua link cisco huawei

```
configure terminal
interface gi0/3
 no shutdown
exit
```

```
interface gi0/0
 no shutdown
exit

show ip route 192.168.50.0
trace 192.168.50.10
```




**Gambar 29:** Hasil trace route setelah dua link utama kembali diaktifkan

## 2 Analisis Hasil Percobaan

Pada bagian pertama praktikum, kita berhasil membagi satu perangkat switch fisik menjadi beberapa jaringan terpisah menggunakan VLAN. Pembagian ini membuat komputer di bagian Finance, HR, IT, dan Marketing memiliki ruang jaringannya sendiri sehingga arus data tidak saling bercampur dan lebih aman. Untuk menghubungkan semua jaringan terpisah tersebut ke sebuah router pengatur lalu lintas, kita menggunakan pengaturan port trunk pada kabel penghubungnya. Dengan jalur trunk ini, router bisa mengenali dan melayani komunikasi antar komputer dari berbagai VLAN hanya melalui satu sambungan kabel fisik saja.

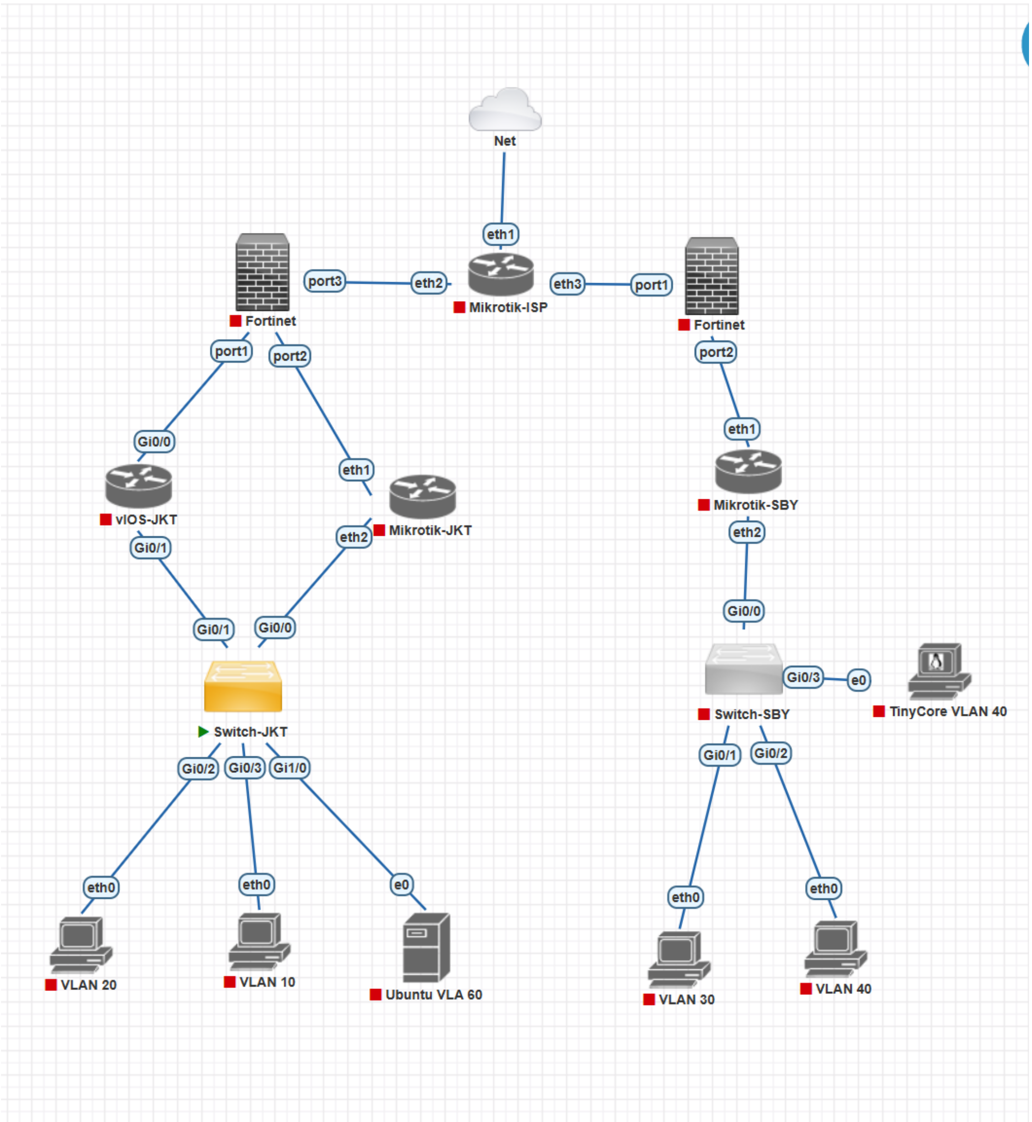
Pada bagian selanjutnya, kita menggunakan OSPF agar jaringan bisa dibuat secara otomatis tanpa harus mengatur rute satu per satu secara manual. OSPF di sini bertindak sebagai bahasa universal yang memungkinkan perangkat dari merk yang berbeda (multivendor), yaitu Cisco, Huawei, dan MikroTik, bisa saling berkenalan dan bertukar informasi rute. Keberhasilan pengujian antar jaringan membuktikan bahwa perbedaan merk perangkat keras bukan sebuah halangan, karena semuanya patuh pada aturan standar OSPF untuk saling menunjukkan jalan menuju komputer tujuan.

Kegunaan OSPF juga saat kita melakukan uji coba pemutusan kabel untuk melihat bagaimana jaringan bertahan dari kerusakan. Saat kabel jalur utama yang menghubungkan Cisco dan Huawei dengan Mikrotik sengaja dimatikan, OSPF secara otomatis menyadari bahwa jalan tersebut buntu dan langsung mengalihkan lalu lintas data ke jalan cadangan yang melewati router satunya. Ketika kabel jalur utama kembali diperbaiki dan dihidupkan, OSPF dengan pintar memindahkan kembali arus data ke jalan semula karena dinilai lebih dekat dan cepat, sehingga jaringan tetap hidup tanpa harus diatur ulang.

Pada challenge, jaringan berhasil disempurnakan secara utuh dengan penambahan FortiGate untuk firewall dan Cloud sebagai gerbang menuju internet yang dihubungkan melalui router MikroTik, yang dibuktikan dengan ping 8.8.8.8 dan google.com dari VPC. Challenge selanjutnya sukses menerapkan pengaturan rute manual (manual routing) untuk memecah arah perjalanan data, sehingga saat dilacak (trace), lalu lintas dari komputer VLAN 10 dan 20 terbukti diarahkan melewati jalan yang berbeda dengan arus dari komputer VLAN 30 dan 40. Keberhasilan memisahkan jalur ini membuktikan bahwa kita tidak hanya mampu membangun jaringan yang terhubung ke internet, tetapi juga dalam memanajemen rute data agar setiap departemen memiliki rutenya masing-masing secara rapi tanpa saling bertumpuk.

## 3 Hasil Tugas Modul

Tugas Modul 5 ini mensimulasikan arsitektur jaringan enterprise yang menghubungkan Kantor Pusat (HQ) di Jakarta dengan Kantor Cabang di Surabaya melalui infrastruktur ISP. Konfigurasi ini mengintegrasikan berbagai teknologi jaringan fundamental, termasuk redundansi gateway menggunakan VRRP, manajemen IP terpusat dengan DHCP Relay pada Ubuntu Server, serta pengamanan jaringan menggunakan firewall FortiGate. Selain itu, komunikasi antar-kantor dirutekan secara dinamis menggunakan protokol OSPF yang berjalan di atas antarmuka GRE Tunnel, memungkinkan kedua situs untuk bertukar informasi secara otomatis seolah berada pada satu lingkup jaringan lokal yang sama.




**Gambar 30:** Topologi Tugas Modul P5

## 4 Kesimpulan

Secara keseluruhan, praktikum ini membuktikan keberhasilan dalam membangun jaringan skala perusahaan yang modern, aman, dan tangguh (ada failover). Fondasi jaringan dibangun menggunakan VLAN dan Trunk untuk mengisolasi lalu lintas data antar departemen secara rapi, yang kemudian disatukan melintasi perangkat beda merk (Cisco, Huawei, MikroTik) menggunakan protokol OSPF, sebuah sistem yang mampu membuat rute secara otomatis dan langsung mengalihkan arus data ke jalur cadangan ketika kabel utama terputus. Infrastruktur ini semakin disempurnakan pada tahap akhir dengan menambahkan FortiGate sebagai keamanan (firewall) dan akses ke Cloud (internet), serta penerapan pengaturan rute manual (manual routing) yang secara sukses memisahkan jalur data agar kelompok VLAN 10 dan 20 melewati jalan yang berbeda dengan VLAN 30 dan 40.

# 5 Lampiran

## 5.1 Dokumentasi saat praktikum